

CompSci 234/NetSys 210
Advanced Topics in Networking
Winter 2019

Unit 10: Software-Defined Networks

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Slide adopted from Prof. Yeh's materials

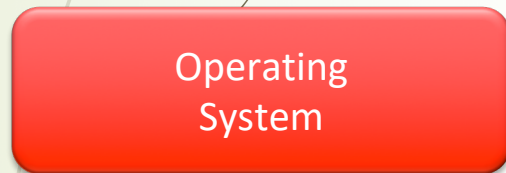
Agenda

- Motivations
- Concepts
- OpenFlow Protocol
- Network Virtualization
- Virtual Switches

Network Switches are Too Complex to Work with...



Routing, management, mobility management, access control, VPNs, ...



Million of lines
of source code

8521 RFCs

Barrier to entry

500M gates
10Gbytes RAM

Bloated

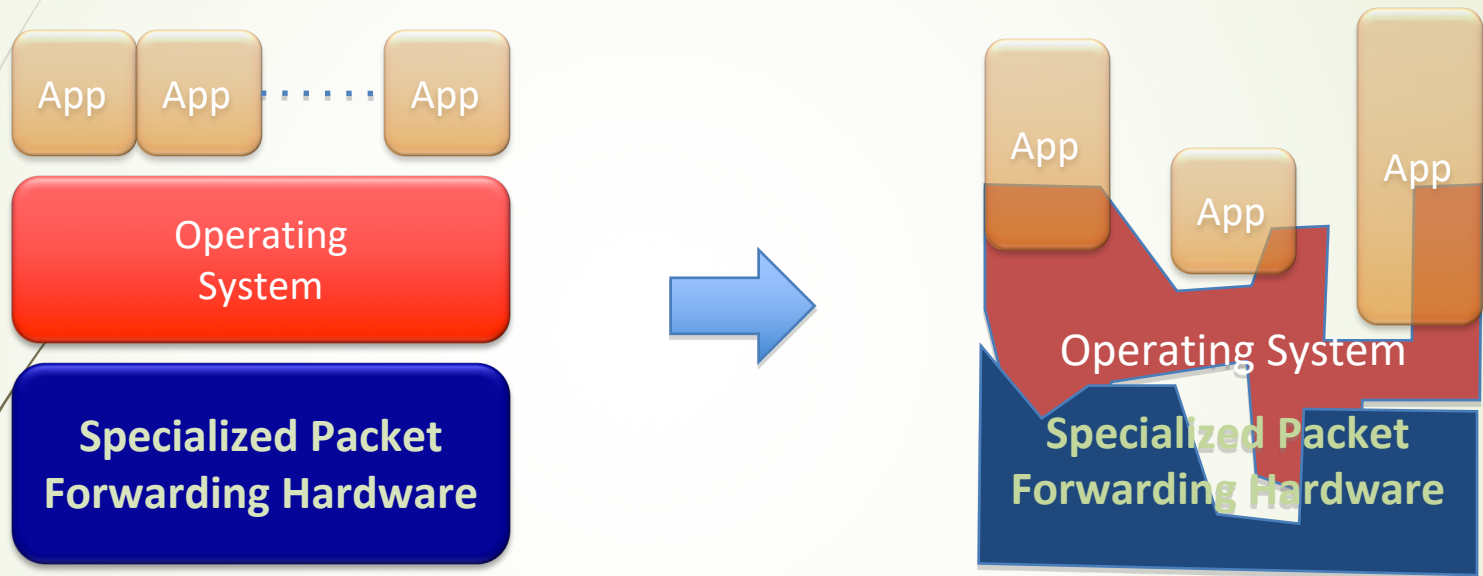
Power Hungry

Many complex functions baked into the infrastructure

*OSPF, BGP, multicast, differentiated services,
Traffic Engineering, NAT, firewalls, MPLS, redundant layers, ...*

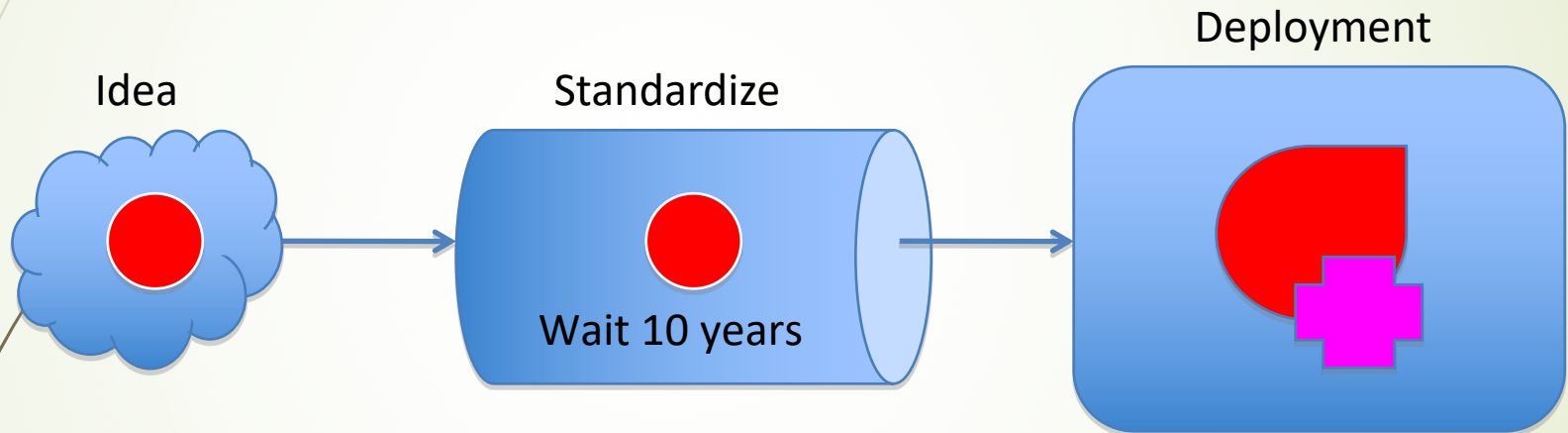
An industry with a “mainframe-mentality”

In Reality, Network Switches Look like This



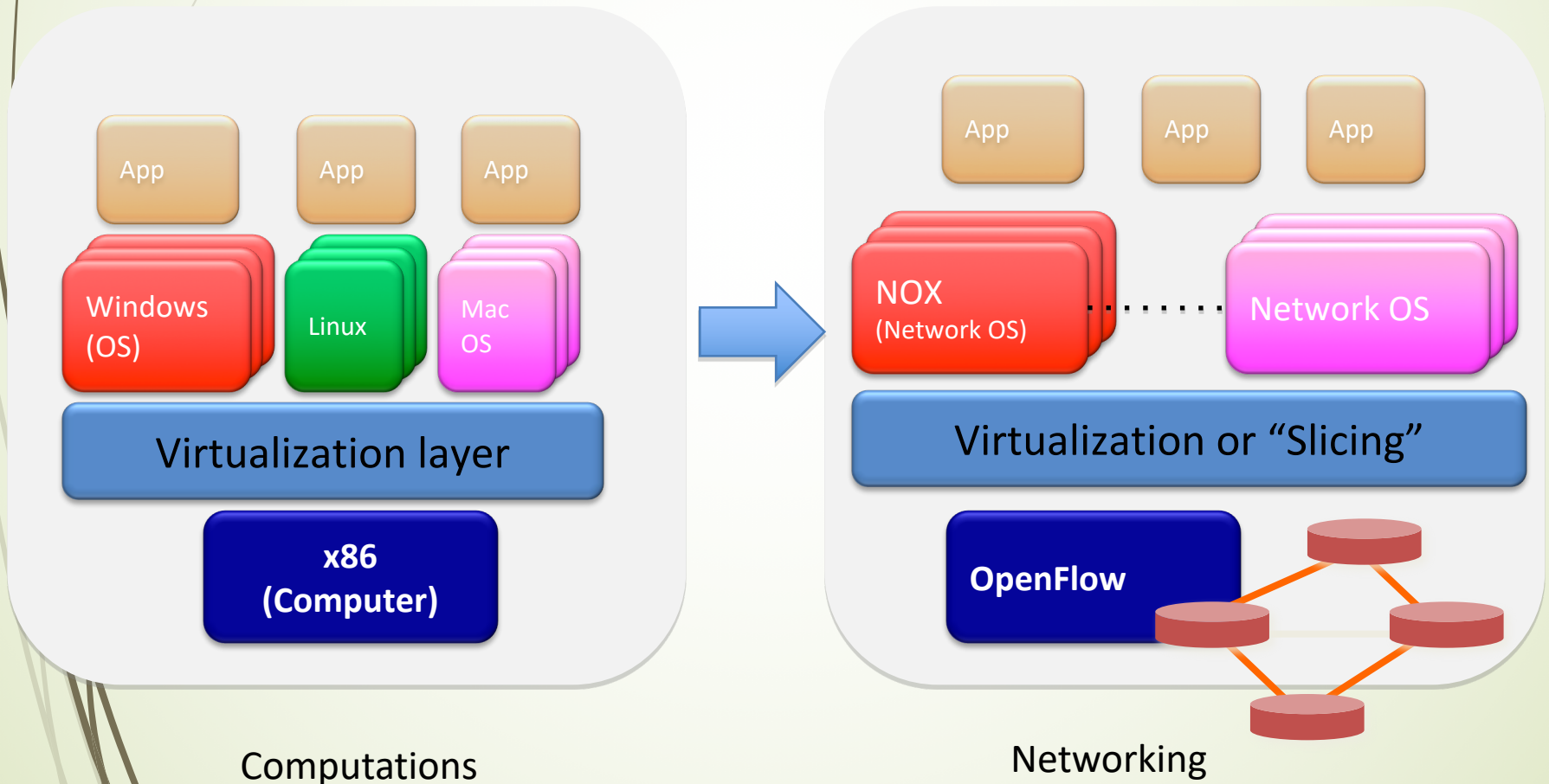
- Lack of competition means slow innovation
- Closed architecture means blurry, closed interfaces
- Vertically integrated, complex, closed, and proprietary
- Not suitable for experimental ideas (bad for researchers)
- Not good for network admins and users

Standardization Process Makes it Even Worse

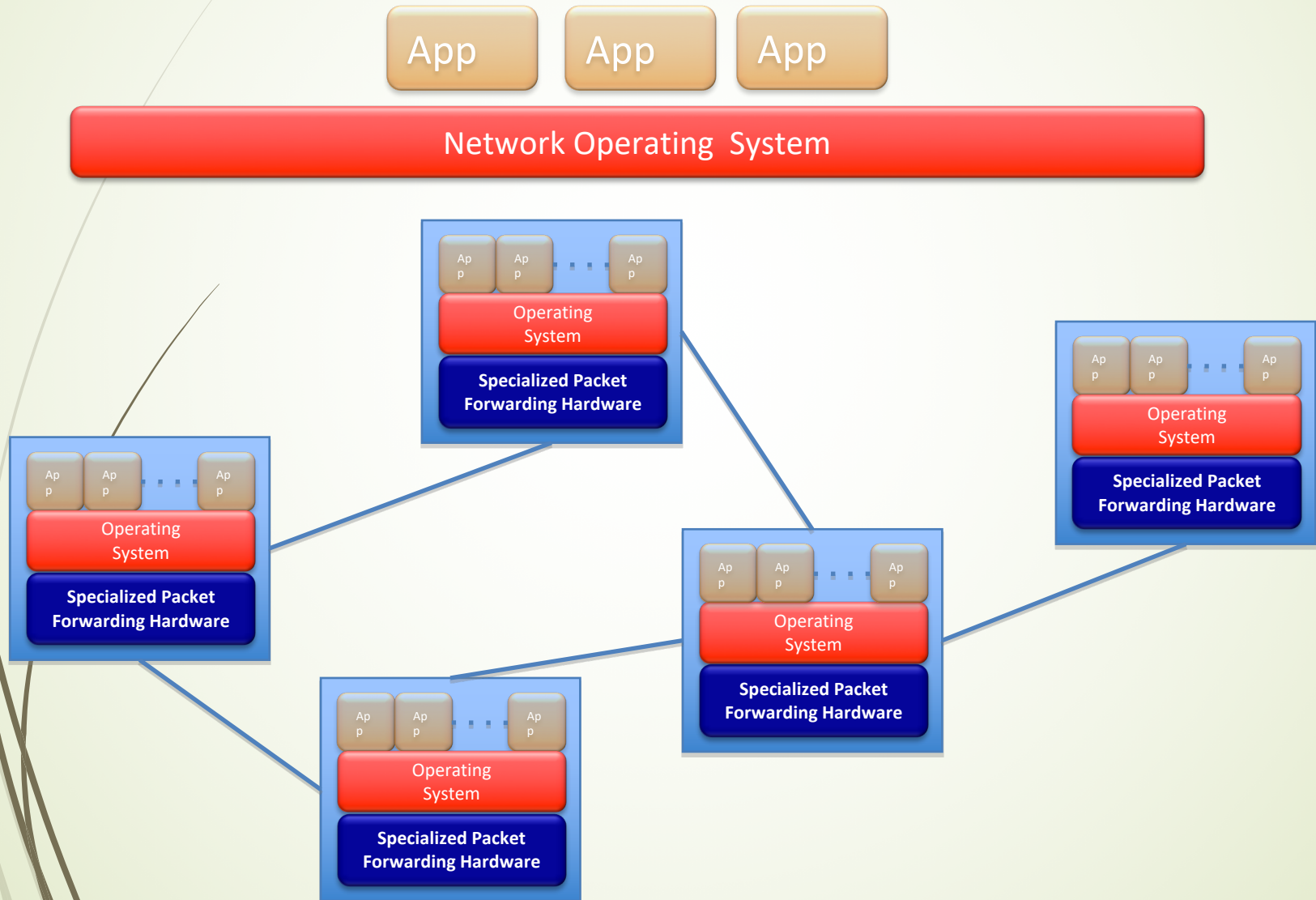


- Driven by vendors
- Consumers largely locked out
- Least intersected denominator features
- Slow innovation

Virtualization for Computations and Networking



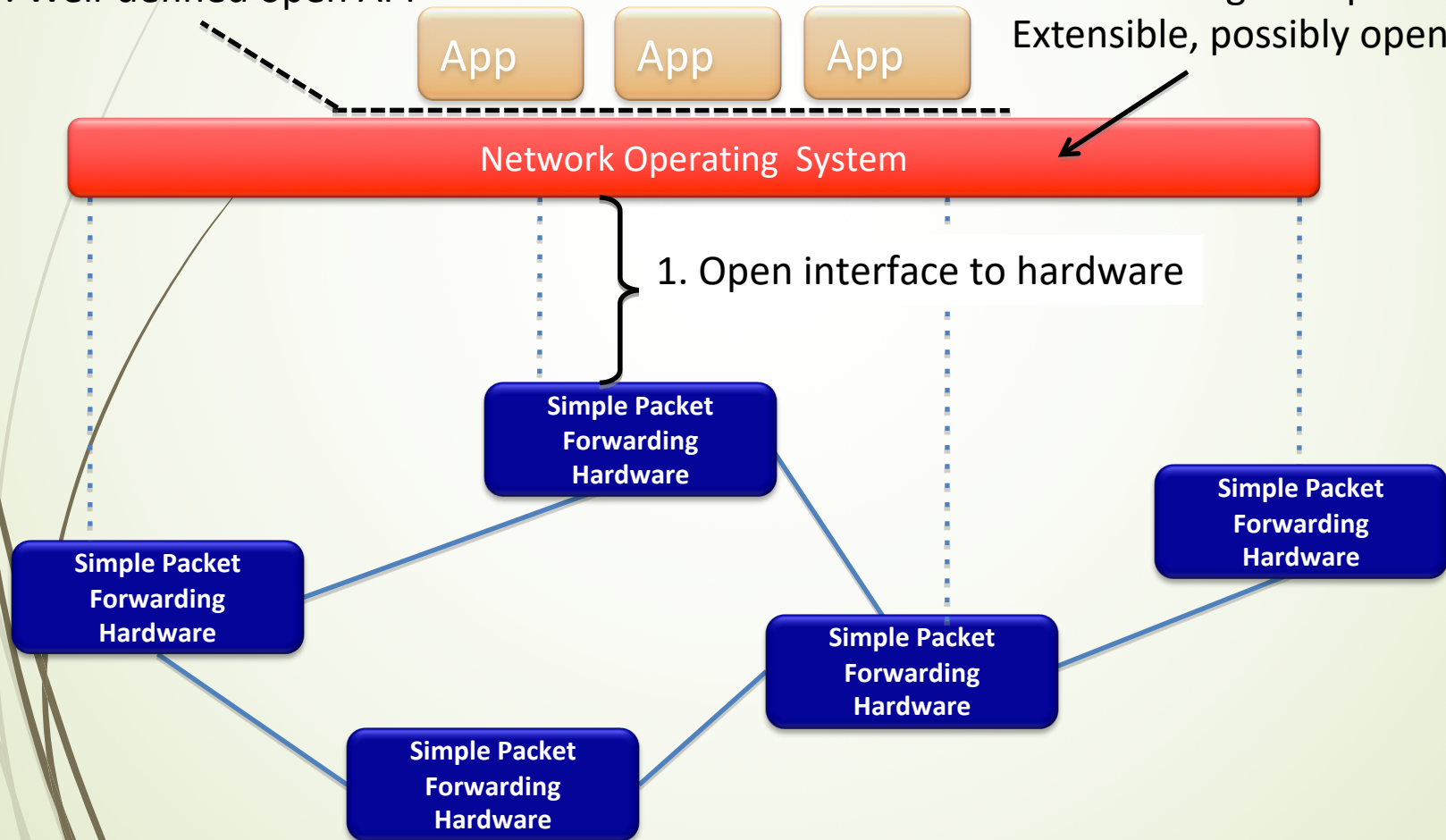
From Today's Network Switches



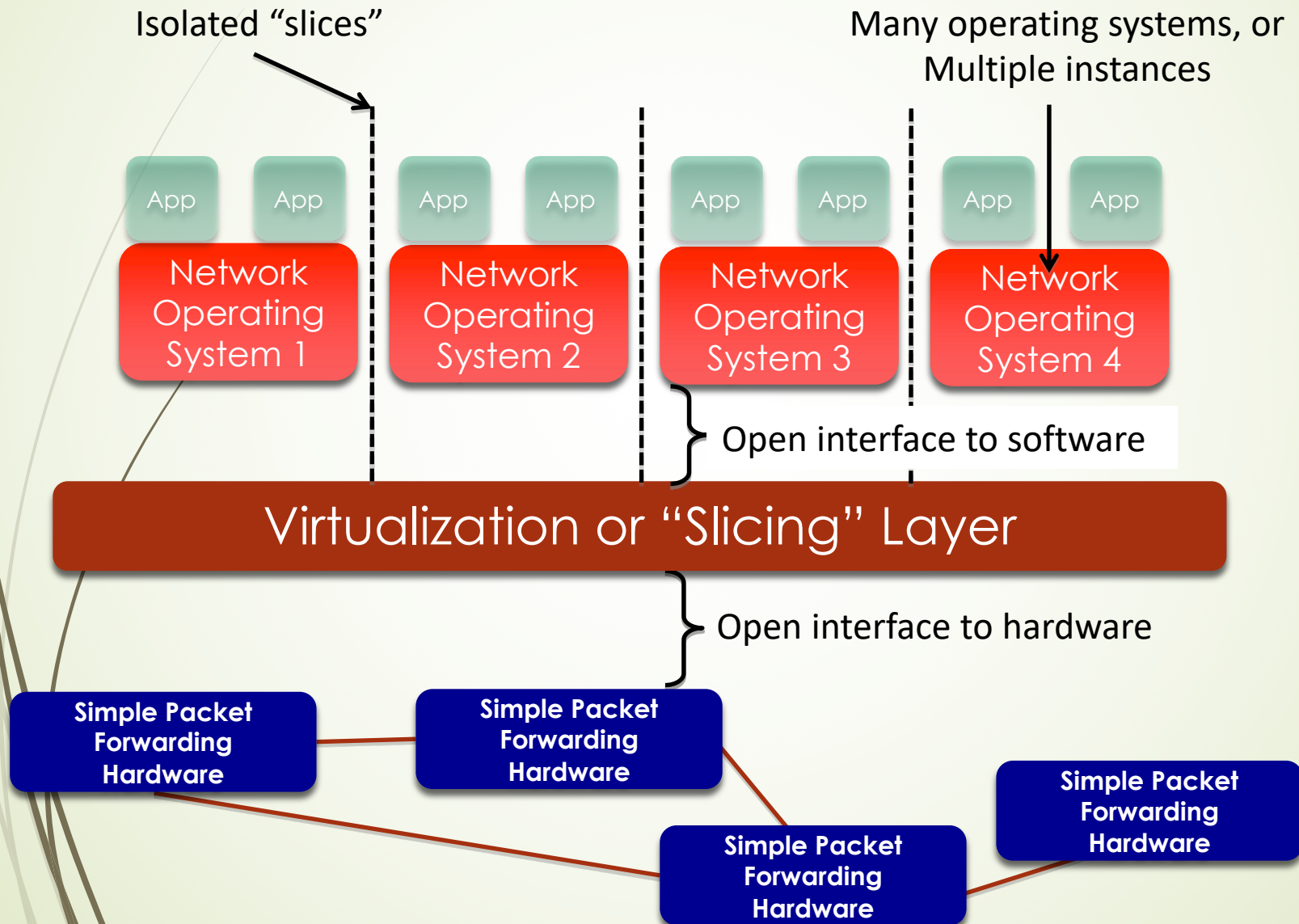
To Software-Defined Networking Switches

3. Well-defined open API

2. At least one good operating system
Extensible, possibly open-source



Network Virtualization Becomes a Reality



Advantages of Software-Defined Networking

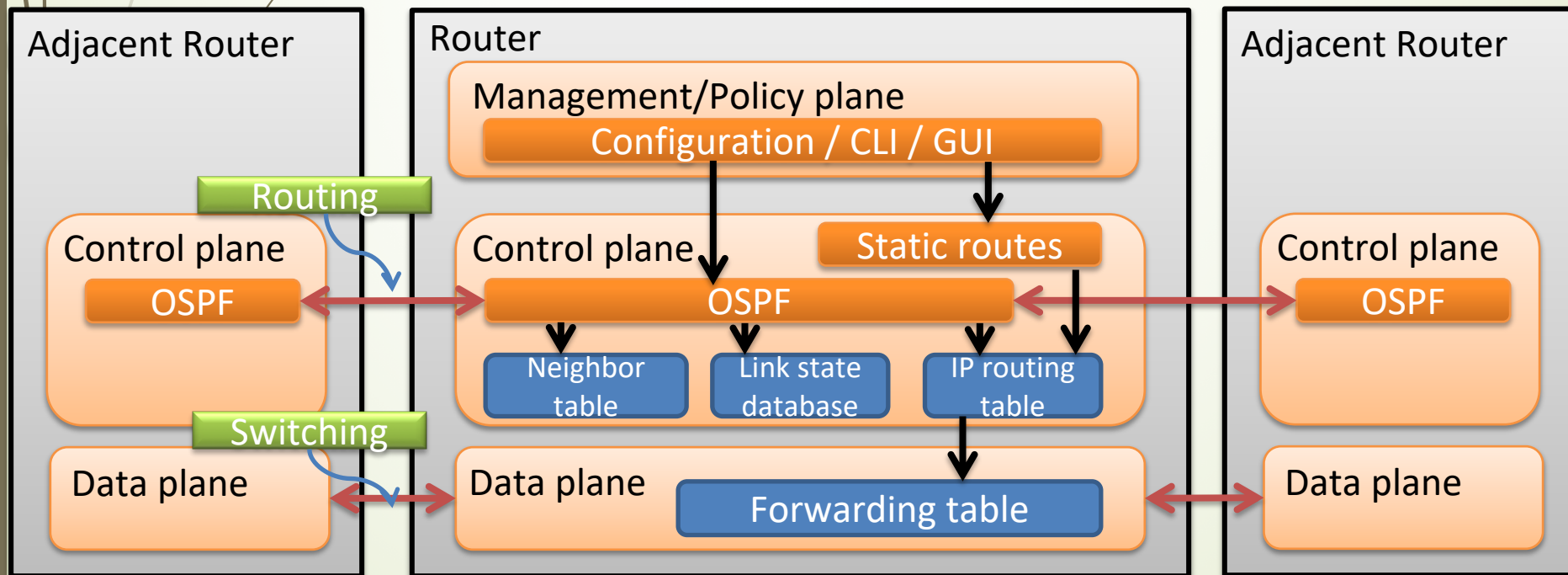
- More innovation in network services
 - Owners, operators, 3rd party developers, researchers can improve the network
 - E.g. energy management, data center management, policy routing, access control, denial of service, mobility
- Lower barrier to entry for competition
 - Healthier markets for new players
- **Anything else?**
- **Q: What are the cons?**
 - Costs ← like other s/w, co-processors, FPGA, ASIC examples! ← Why we have to learn P4 two days later....
 - Legacy networks/switches/hosts
 - New applications (more precisely, which cannot be realized before SDN appears?)

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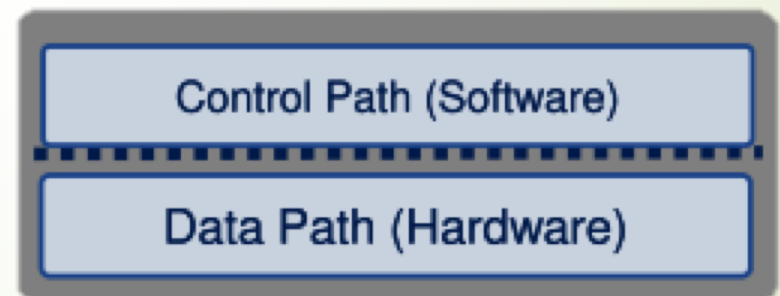
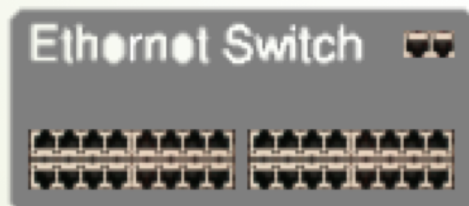
Traditional Routers are Partitioned into...

- Management plane/ configuration
- **Control plane** / Decision: OSPF
- **Data plane** / Forwarding: Longest Prefix Matching



Traditional Switches are Partitioned into...

- **Software:** Control Plane – The brain and decision maker
- **Hardware:** Data Plane – Packet forwarder



Key Concepts of SDN

- Separate Control plane and Data plane entities
 - Network intelligence and state are logically centralized
 - The underlying network infrastructure is abstracted from the applications
- Execute or run Control plane software on general purpose hardware
 - Decouple from specific networking hardware
 - Use commodity servers for routing
- Have programmable data planes
 - Maintain, control and program data plane state from a central entity
- An architecture to control not just a networking device but an entire network